

Unit I – Laplace Transform

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1 Definition of the Laplace Transform

Definition: Laplace Transform

The Laplace transform of any continuous function $f(t)$ is defined by

$$\mathcal{L}(f(t)) = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

Here s may be a real or complex number.

2 Existence of the Laplace Transform

Existence of L.T.

The Laplace transform of a function $f(t)$ exists if

- (i) $f(t)$ must be a continuous or piece-wise continuous function and has a finite number of discontinuities;
- (ii) $\lim_{t \rightarrow \infty} f(t) e^{-st}$ is finite, i.e. $|f(t)| \leq M e^{at}$ (exponential order).

Examples where the L.T. does not exist: (a) $\tan t$ (b) $\frac{1 - e^t}{t}$

3 Laplace Transform of Some Basic Functions

(1) $\mathcal{L}(e^{at})$

By definition:

$$\begin{aligned} \mathcal{L}(f(t)) &= \int_0^{\infty} e^{-st} f(t) dt \\ \Rightarrow \mathcal{L}(e^{at}) &= \int_0^{\infty} e^{-st} e^{at} dt \\ &= \int_0^{\infty} e^{-(s-a)t} dt \\ &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} \quad \left(\because \int e^{ax} dx = \frac{e^{ax}}{a} \right) \\ &= -\frac{1}{s-a} (e^{-\infty} - e^0), \quad \text{provided } s-a > 0 \end{aligned}$$

$$= -\frac{1}{s-a} (0-1), \quad \text{provided } s > a$$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}, \quad s > a \quad \text{and similarly} \quad \mathcal{L}(e^{-at}) = \frac{1}{s+a}$$

(2) $\mathcal{L}(\sin at)$

We know that

$$\sin at = \frac{e^{iat} - e^{-iat}}{2i} \quad \left(\because \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}, \quad \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \right)$$

Therefore,

$$\begin{aligned} \Rightarrow \mathcal{L}(\sin at) &= \frac{1}{2i} \mathcal{L}(e^{iat} - e^{-iat}) \\ &= \frac{1}{2i} [\mathcal{L}(e^{iat}) - \mathcal{L}(e^{-iat})] \\ &= \frac{1}{2i} \left[\frac{1}{s-ia} - \frac{1}{s+ia} \right] \quad \left(\because \mathcal{L}(e^{at}) = \frac{1}{s-a} \right) \\ &= \frac{1}{2i} \left[\frac{(s+ia) - (s-ia)}{(s-ia)(s+ia)} \right] \\ &= \frac{1}{2i} \left[\frac{2ia}{s^2 - (ia)^2} \right] \end{aligned}$$

$$\mathcal{L}(\sin at) = \frac{a}{s^2 + a^2} \quad (\because i^2 = -1)$$

Similarly, using $\cos at = \frac{e^{iat} + e^{-iat}}{2}$ we obtain

$$\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$$

(3) $\mathcal{L}(\sinh at)$

We know that

$$\sinh at = \frac{e^{at} - e^{-at}}{2}$$

Therefore,

$$\begin{aligned} \Rightarrow \mathcal{L}(\sinh at) &= \frac{1}{2} \mathcal{L}(e^{at} - e^{-at}) \\ &= \frac{1}{2} [\mathcal{L}(e^{at}) - \mathcal{L}(e^{-at})] \\ &= \frac{1}{2} \left[\frac{1}{s-a} - \frac{1}{s+a} \right] = \frac{1}{2} \left[\frac{(s+a) - (s-a)}{(s-a)(s+a)} \right] \\ &= \frac{1}{2} \left[\frac{2a}{s^2 - a^2} \right] \end{aligned}$$

$$\mathcal{L}(\sinh at) = \frac{a}{s^2 - a^2}$$

(4) $\mathcal{L}(\cosh at)$

$$\mathcal{L}(\cosh at) = \frac{s}{s^2 - a^2}$$

We know that $\cosh at = \frac{e^{at} + e^{-at}}{2}$; proceeding exactly as above we get the answer.

(5) $\mathcal{L}(t^n)$

$$\mathcal{L}(t^n) = \frac{n!}{s^{n+1}} \quad \text{or} \quad \frac{\Gamma(n+1)}{s^{n+1}}$$

Proof: We know that

$$\begin{aligned} \Gamma(n) &= \int_0^\infty e^{-x} x^{n-1} dx \\ \Rightarrow \Gamma(n) &= \int_0^\infty e^{-t} t^{n-1} dt \quad (\text{replacing } x \text{ by } t) \\ \Rightarrow \frac{\Gamma(n)}{a^n} &= \int_0^\infty e^{-at} t^{n-1} dt \quad (\text{replacing } t \text{ by } at) \\ \Rightarrow \int_0^\infty e^{-st} t^{(n+1)-1} dt &= \frac{\Gamma(n+1)}{s^{n+1}} \quad (\text{taking } a = s, n \rightarrow n+1) \\ \Rightarrow \int_0^\infty e^{-st} t^n dt &= \frac{\Gamma(n+1)}{s^{n+1}} \\ \Rightarrow \mathcal{L}(t^n) &= \frac{\Gamma(n+1)}{s^{n+1}} = \frac{n!}{s^{n+1}} \quad (\text{if } n \text{ is a +ve integer}) \end{aligned}$$

Deductions**Put $n = \frac{1}{2}$ in the above result:**

$$\begin{aligned} \text{i.e. } \mathcal{L}(t^n) &= \frac{\Gamma(n+1)}{s^{n+1}} \\ \Rightarrow \mathcal{L}(t^{1/2}) &= \frac{\Gamma(\frac{1}{2}+1)}{s^{1/2+1}} = \frac{\frac{1}{2}\Gamma(\frac{1}{2})}{s^{3/2}} \quad (\because \Gamma(n+1) = n\Gamma(n)) \end{aligned}$$

$$\mathcal{L}(\sqrt{t}) = \frac{\sqrt{\pi}}{2s^{3/2}} \quad (\because \Gamma(\frac{1}{2}) = \sqrt{\pi})$$

Put $n = -\frac{1}{2}$:

$$\mathcal{L}(t^{-1/2}) = \frac{\Gamma(-\frac{1}{2}+1)}{s^{-1/2+1}} = \frac{\Gamma(\frac{1}{2})}{s^{1/2}} = \frac{\sqrt{\pi}}{\sqrt{s}}$$

$$\Rightarrow \mathcal{L}\left(\frac{1}{\sqrt{t}}\right) = \sqrt{\frac{\pi}{s}}$$

Put $n = 1$ in $\mathcal{L}(t^n) = \frac{n!}{s^{n+1}}$:

$$\Rightarrow \mathcal{L}(t) = \frac{1!}{s^2} = \frac{1}{s^2}$$

Similarly,

$$\mathcal{L}(t^2) = \frac{2}{s^3}, \quad \mathcal{L}(t^3) = \frac{6}{s^4} = \frac{3!}{s^4}.$$

4 Properties of the Laplace Transform

4.1 First Shifting Property

Property 1: First Shifting Property

If $\mathcal{L}(f(t)) = F(s)$ then $\mathcal{L}(e^{at}f(t)) = F(s - a)$.

Proof: By definition,

$$\begin{aligned}\mathcal{L}(f(t)) &= \int_0^\infty e^{-st} f(t) dt = F(s) \\ \therefore \mathcal{L}(e^{at}f(t)) &= \int_0^\infty e^{-st} e^{at} f(t) dt \\ &= \int_0^\infty e^{-(s-a)t} f(t) dt \\ \therefore \mathcal{L}(e^{at}f(t)) &= F(s - a) \quad (\because \text{by definition})\end{aligned}$$

By the above result,

$$\mathcal{L}(e^{at}f(t)) = (F(s))_{s \rightarrow s-a}$$

$$\mathcal{L}(e^{at}f(t)) = \left(\mathcal{L}(f(t)) \right)_{s \rightarrow s-a}$$

4.2 Change of Scale Property

Property 2: Change of Scale Property

If $\mathcal{L}(f(t)) = F(s)$ then $\mathcal{L}(f(at)) = \frac{1}{a} F\left(\frac{s}{a}\right)$.

Proof: By definition,

$$\begin{aligned}\mathcal{L}(f(t)) &= \int_0^\infty e^{-st} f(t) dt = F(s) \\ \therefore \mathcal{L}(f(at)) &= \int_0^\infty e^{-st} f(at) dt\end{aligned} \tag{1}$$

Put $at = u$, so that

$$t = \frac{u}{a} \Rightarrow dt = \frac{1}{a} du, \quad \text{no change in the limits.}$$

$$\begin{aligned}&= \int_0^\infty e^{-s(\frac{u}{a})} f(u) \frac{1}{a} du \\ &= \frac{1}{a} \int_0^\infty e^{-\left(\frac{s}{a}\right)u} f(u) du \\ &= \frac{1}{a} F\left(\frac{s}{a}\right) \quad (\because \text{by definition}) \\ \text{i.e. } \mathcal{L}(f(at)) &= \frac{1}{a} F\left(\frac{s}{a}\right)\end{aligned}$$

Similarly,

$$\mathcal{L}\left(f\left(\frac{t}{a}\right)\right) = a F(as)$$

Problem: Find the L.T. of $e^{-3t} \sin 2t$

Solution:

$$\begin{aligned}
 \mathcal{L}(e^{-3t} \sin 2t) &= [\mathcal{L}(\sin 2t)]_{s \rightarrow s+3} \quad (\because \text{by first shifting property}) \\
 &= \left[\frac{2}{s^2 + 2^2} \right]_{s \rightarrow s+3} \\
 &= \frac{2}{(s+3)^2 + 4} \\
 &= \frac{2}{s^2 + 6s + 9 + 4} = \frac{2}{s^2 + 6s + 13} //
 \end{aligned}$$

5 Laplace Transform of Periodic Functions

Periodic Function

A function $f(t)$ is said to be periodic if it repeats itself after a certain period a , i.e.

$$f(t+a) = f(t) \quad \text{for all } t.$$

E.g. $\sin t$, $\cos t$, $\tan t$ are periodic functions of period 2π .

Result: L.T. of a periodic function

If $f(t)$ is a periodic function with period a , i.e. $f(t+a) = f(t)$, then

$$\mathcal{L}(f(t)) = \frac{1}{1 - e^{-as}} \int_0^a e^{-st} f(t) dt$$

Problem 1: Sine-wave rectifier function

Find the L.T. of the sine wave rectifier function

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases} \quad \text{of period } \frac{2\pi}{\omega}.$$

Solution: Given $f(t)$ is a periodic function with period $\frac{2\pi}{\omega}$

$$\Rightarrow a = \frac{2\pi}{\omega}$$

$$\begin{aligned}
 \therefore \mathcal{L}(f(t)) &= \frac{1}{1 - e^{-as}} \int_0^a e^{-st} f(t) dt \\
 &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \int_0^{2\pi/\omega} e^{-st} f(t) dt \quad \left(\because a = \frac{2\pi}{\omega} \right) \\
 &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \left[\left(\int_0^{\pi/\omega} + \int_{\pi/\omega}^{2\pi/\omega} \right) e^{-st} f(t) dt \right] \\
 &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \left[\int_0^{\pi/\omega} e^{-st} \sin \omega t dt + \int_{\pi/\omega}^{2\pi/\omega} e^{-st} \cdot 0 dt \right]
 \end{aligned}$$

$$= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \int_0^{\pi/\omega} e^{-st} \sin \omega t \, dt$$

Using the standard integral

$$\begin{aligned} \int e^{ax} \sin bx \, dx &= \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx), \\ &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \left[\frac{e^{-st}}{s^2 + \omega^2} (-s \sin \omega t - \omega \cos \omega t) \right]_0^{\pi/\omega} \\ &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \left[\frac{e^{-s\pi/\omega}}{s^2 + \omega^2} (0 - \omega(-1)) - \frac{1}{s^2 + \omega^2} (0 - \omega(1)) \right] \\ &= \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} \left[\frac{\omega e^{-s\pi/\omega}}{s^2 + \omega^2} + \frac{\omega}{s^2 + \omega^2} \right] \\ &= \frac{\omega}{s^2 + \omega^2} \cdot \frac{1}{1 - e^{-\frac{2\pi}{\omega}s}} (1 + e^{-s\pi/\omega}) \\ &= \frac{\omega}{s^2 + \omega^2} \cdot \frac{1 + e^{-\pi s/\omega}}{(1 + e^{-\pi s/\omega})(1 - e^{-\pi s/\omega})} \end{aligned}$$

$$\mathcal{L}(f(t)) = \frac{\omega}{(s^2 + \omega^2)(1 - e^{-\pi s/\omega})} //$$

Problem 2: Triangular wave function

Find the L.T. of the triangular wave function

$$f(t) = \begin{cases} t, & 0 \leq t \leq a \\ 2a - t, & a \leq t \leq 2a \end{cases} \quad \text{of period } 2a.$$

Solution: Given $f(t)$ is a periodic function with period $2a$.

$$\begin{aligned} \therefore \mathcal{L}(f(t)) &= \frac{1}{1 - e^{-as}} \int_0^a e^{-st} f(t) \, dt \\ &= \frac{1}{1 - e^{-2as}} \int_0^{2a} e^{-st} f(t) \, dt \quad (\because \text{period } a = 2a) \\ &= \frac{1}{1 - e^{-2as}} \left[\left(\int_0^a + \int_a^{2a} \right) e^{-st} f(t) \, dt \right] \\ &= \frac{1}{1 - e^{-2as}} \left[\int_0^a e^{-st} t \, dt + \int_a^{2a} e^{-st} (2a - t) \, dt \right] \end{aligned}$$

Using Bernoulli's formula $\int u v \, dt = uv_1 - u'v_2 + u''v_3 - \dots$

$$\begin{aligned} &= \frac{1}{1 - e^{-2as}} \left\{ \left[t \left(\frac{e^{-st}}{-s} \right) - 1 \left(\frac{e^{-st}}{(-s)^2} \right) + 0 \right]_0^a + \left[(2a - t) \left(\frac{e^{-st}}{-s} \right) - (-1) \left(\frac{e^{-st}}{(-s)^2} \right) + 0 \right]_a^{2a} \right\} \\ &= \frac{1}{1 - e^{-2as}} \left\{ -\frac{1}{s} \left(t e^{-st} \right)_0^a - \frac{1}{s^2} \left(e^{-st} \right)_0^a - \frac{1}{s} \left((2a - t) e^{-st} \right)_a^{2a} + \frac{1}{s^2} \left(e^{-st} \right)_a^{2a} \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{1 - e^{-2as}} \left\{ -\frac{1}{s}(ae^{-as} - 0) - \frac{1}{s^2}(e^{-as} - 1) - \frac{1}{s}(0 - ae^{-as}) + \frac{1}{s^2}(e^{-2as} - e^{-as}) \right\} \\
&= \frac{1}{1 - e^{-2as}} \left\{ \frac{1}{s^2}(e^{-2as} - e^{-as} - e^{-as} + 1) \right\} \\
&= \frac{1}{1 - e^{-2as}} \left\{ \frac{1}{s^2}(e^{-2as} - 2e^{-as} + 1) \right\} \\
&= \frac{1}{1 - e^{-2as}} \left\{ \frac{1}{s^2}(e^{-as} - 1)^2 \right\} \\
&= \frac{1}{(1 + e^{-as})(1 - e^{-as})} \cdot \frac{1}{s^2} (1 - e^{-as})^2 \\
&= \frac{1}{s^2} \left(\frac{1 - e^{-as}}{1 + e^{-as}} \right) \\
&= \frac{1}{s^2} \left(\frac{1 - e^{-\frac{as}{2}} e^{-\frac{as}{2}}}{1 + e^{-\frac{as}{2}} e^{-\frac{as}{2}}} \right) = \frac{1}{s^2} \left(\frac{1 - \frac{e^{-as/2}}{e^{as/2}}}{1 + \frac{e^{-as/2}}{e^{as/2}}} \right) \\
&= \frac{1}{s^2} \left(\frac{(e^{as/2} - e^{-as/2})/e^{as/2}}{(e^{as/2} + e^{-as/2})/e^{as/2}} \right) \quad \left(\because \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \right)
\end{aligned}$$

$$\mathcal{L}(f(t)) = \frac{1}{s^2} \tanh\left(\frac{as}{2}\right) //$$

Exercise: Square wave function

Find the L.T. of

$$f(t) = \begin{cases} k, & 0 \leq t \leq a \\ -k, & a \leq t \leq 2a \end{cases} \quad \text{of period } 2a.$$

6 Laplace Transform of Derivatives

Property: L.T. of derivatives

If $\mathcal{L}(f(t)) = F(s)$, then $\mathcal{L}(f'(t)) = sF(s) - f(0)$.

Proof: By definition,

$$\begin{aligned}
\mathcal{L}(f(t)) &= \int_0^\infty e^{-st} f(t) dt = F(s) \\
\Rightarrow \mathcal{L}(f'(t)) &= \int_0^\infty e^{-st} f'(t) dt \\
&= \int_0^\infty e^{-st} d \underset{dv}{f(t)} \quad \left(\begin{array}{l} u = e^{-st} \\ du = e^{-st}(-s) dt \end{array} \right) \\
&= uv - \int v du \\
&= \left(e^{-st} f(t) \right)_0^\infty - \int_0^\infty f(t) e^{-st}(-s) dt
\end{aligned}$$

$$\begin{aligned}
&= [0 - f(0)] + s \int_0^{\infty} f(t) e^{-st} dt \\
&= -f(0) + s F(s)
\end{aligned}$$

$$\Rightarrow \mathcal{L}(f'(t)) = sF(s) - f(0)$$

Similarly,

$$\begin{aligned}
\mathcal{L}(f''(t)) &= s^2 F(s) - sf(0) - f'(0) \\
\mathcal{L}(f'''(t)) &= s^3 F(s) - s^2 f(0) - sf'(0) - f''(0)
\end{aligned}$$

and so on.

7 Laplace Transform of Integrals

Result: L.T. of integrals

If $\mathcal{L}(f(t)) = F(s)$ then

$$\mathcal{L}\left(\int_0^t f(t) dt\right) = \frac{F(s)}{s}$$

Problems: Find the L.T. of the following

$$(1) \int_0^t \frac{\sin t}{t} dt \quad (2) \int_0^t \frac{e^t \sin t}{t} dt \quad (3) \int_0^t e^{-t} \cos t dt$$

Solution (1):

$$\begin{aligned}
\mathcal{L}\left(\int_0^t \frac{\sin t}{t} dt\right) &= \mathcal{L}\left(\int_0^t f(t) dt\right) \quad \left(\text{where } f(t) = \frac{\sin t}{t}\right) \\
&= \frac{F(s)}{s} \\
&= \frac{1}{s} \mathcal{L}(f(t)) \\
&= \frac{1}{s} \mathcal{L}\left(\frac{\sin t}{t}\right) \\
&= \frac{1}{s} \left[\int_s^{\infty} \mathcal{L}(\sin t) ds \right] \quad (\because \text{division by } t) \\
&= \frac{1}{s} \int_s^{\infty} \frac{1}{s^2 + 1} ds \\
&= \frac{1}{s} \left(\tan^{-1}(s) \right)_s^{\infty} \\
&= \frac{1}{s} \left(\tan^{-1}(\infty) - \tan^{-1}(s) \right)
\end{aligned}$$

$$\mathcal{L}\left(\int_0^t \frac{\sin t}{t} dt\right) = \frac{1}{s} \left(\frac{\pi}{2} - \tan^{-1}(s) \right)$$

Solution (2):

$$\mathcal{L}\left(\int_0^t \frac{e^t \sin t}{t} dt\right) = \frac{1}{s} \mathcal{L}\left(\frac{e^t \sin t}{t}\right) \quad \left(\because \mathcal{L} \int_0^t f(t) dt = \frac{F(s)}{s}\right)$$

$$\begin{aligned}
&= \frac{1}{s} \left[\int_s^\infty \mathcal{L}(e^t \sin t) ds \right] \quad (\because \text{division by } t) \\
&= \frac{1}{s} \left[\int_s^\infty (\mathcal{L}(\sin t))_{s \rightarrow s-1} ds \right] \\
&= \frac{1}{s} \int_s^\infty \left(\frac{1}{s^2 + 1} \right)_{s \rightarrow s-1} ds \\
&= \frac{1}{s} \left\{ \int_s^\infty \frac{ds}{(s-1)^2 + 1} \right\} \\
&= \frac{1}{s} \left[\tan^{-1}(s-1) \right]_s^\infty \\
&= \frac{1}{s} \left[\frac{\pi}{2} - \tan^{-1}(s-1) \right]
\end{aligned}$$

$$\mathcal{L}\left(\frac{e^t \sin t}{t}\right) \text{ integrated: } \mathcal{L}\left(\int_0^t \frac{e^t \sin t}{t} dt\right) = \frac{1}{s} \left\{ \frac{\pi}{2} - \tan^{-1}(s-1) \right\} //$$

8 Laplace Transform of Special Functions

8.1 Unit Step Function

Unit Step Function

The unit step function (or unit impulse function) is defined by

$$u_a(t) = u(t-a) = \begin{cases} 1, & t > a > 0 \\ 0, & t < a \end{cases}$$

L.T. of the unit step function

$$\mathcal{L}(u_a(t)) = \mathcal{L}(u(t-a)) = \frac{e^{-as}}{s}$$

Proof: By definition,

$$\begin{aligned}
\mathcal{L}(f(t)) &= \int_0^\infty e^{-st} f(t) dt = F(s) \\
\Rightarrow \mathcal{L}(u_a(t)) &= \left(\int_0^a + \int_a^\infty \right) e^{-st} u_a(t) dt \\
&= 0 + \int_a^\infty e^{-st} \cdot 1 dt \\
&= \int_a^\infty e^{-st} dt = \left(\frac{e^{-st}}{-s} \right)_a^\infty = -\frac{1}{s} (0 - e^{-as})
\end{aligned}$$

$$\mathcal{L}(u_a(t)) = \frac{e^{-as}}{s} = \mathcal{L}(u(t-a)) //$$

8.2 Dirac-Delta Function

Dirac-Delta Function

The Dirac-delta function is denoted by $\delta_a(t)$ or $\delta(t - a)$ and is defined by

$$\delta_a(t) = \lim_{h \rightarrow 0} \begin{cases} \frac{1}{h}, & a < t < a + h, \quad a > 0 \\ 0, & t < a \text{ and } t > a + h \end{cases}$$

L.T. of the Dirac-delta function

By definition,

$$\mathcal{L}(f(t)) = \int_0^\infty e^{-st} f(t) dt$$

$$\begin{aligned} \Rightarrow \mathcal{L}(\delta_a(t)) &= \int_0^\infty e^{-st} \delta_a(t) dt \\ &= \left(\int_0^a + \int_a^{a+h} + \int_{a+h}^\infty \right) e^{-st} \delta_a(t) dt \\ &= 0 + \int_a^{a+h} \lim_{h \rightarrow 0} \frac{1}{h} e^{-st} dt + 0 \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_a^{a+h} e^{-st} dt \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{e^{-st}}{-s} \right)_a^{a+h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(-\frac{1}{s} \right) (e^{-s(a+h)} - e^{-sa}) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(-\frac{1}{s} \right) e^{-as} (e^{-sh} - 1) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \frac{e^{-as}}{s} (1 - e^{-sh}) \\ &= \frac{e^{-as}}{s} \lim_{h \rightarrow 0} \frac{1 - e^{-sh}}{h} \quad \left(= \frac{0}{0} \text{ form; apply L'Hôpital's Rule} \right) \\ &= \frac{e^{-as}}{s} \lim_{h \rightarrow 0} \frac{-e^{-sh}(-s)}{1} \\ &= \frac{e^{-as}}{s} (s) \\ &= e^{-as} \end{aligned}$$

$$\Rightarrow \mathcal{L}(\delta_a(t)) = e^{-as}$$

Note: In the above result, if we put $a = 0$, then

$$\mathcal{L}(\delta_0(t)) = \mathcal{L}(\delta(t)) = 1$$

$$\Rightarrow \mathcal{L}(\delta(t)) = 1$$

8.3 Second Shifting Property

Property: Second Shifting Property

If $\mathcal{L}(f(t)) = F(s)$ then

$$\mathcal{L}(f(t-a)u(t-a)) = e^{-as}F(s)$$

Proof: We know that, by definition,

$$u(t-a) = \begin{cases} 1, & t > a > 0 \\ 0, & t < a \end{cases} \Rightarrow f(t-a)u(t-a) = \begin{cases} f(t-a), & t > a \\ 0, & t < a \end{cases} \quad (a > 0)$$

$$\begin{aligned} \therefore \mathcal{L}(f(t-a)u(t-a)) &= \int_0^{\infty} e^{-st}(f(t-a)u(t-a)) dt \\ &= \int_a^{\infty} e^{-st} f(t-a) dt \end{aligned}$$

Put $u = t - a$, so that

$$t = u + a \Rightarrow dt = du; \quad \text{when } t = a \Rightarrow u = 0, \quad t \rightarrow \infty \Rightarrow u \rightarrow \infty$$

$$\begin{aligned} &= \int_0^{\infty} e^{-s(u+a)} f(u) du \\ &= e^{-as} \int_0^{\infty} e^{-su} f(u) du \\ &= e^{-as} \mathcal{L}(f) \\ &= e^{-as} F(s) \end{aligned}$$

$$\Rightarrow \mathcal{L}(f(t-a)u(t-a)) = e^{-as}F(s)$$

Note: By the above result, we see that

$$\mathcal{L}(f(t-a)u(t-a)) = e^{-as}F(s) = e^{-as} \mathcal{L}(f(t))$$

Problems: Find the L.T. of the following

$$(1) e^{t-1}u(t-1) \quad (2) t^2 u(t-3)$$

Solution (1): By the second shifting property,

$$\mathcal{L}(f(t-a)u(t-a)) = e^{-as} \mathcal{L}(f(t))$$

$$\begin{aligned} \therefore \mathcal{L}(e^{t-1}u(t-1)) &= e^{-s} \mathcal{L}(e^t) \\ &= e^{-s} \frac{1}{s-1} \end{aligned}$$

$$\mathcal{L}(e^{t-1}u(t-1)) = \frac{e^{-s}}{s-1}$$

Solution (2):

$$\begin{aligned}
 \mathcal{L}(t^2 u(t-3)) &= \mathcal{L}(t^2 f(t)) \quad \text{where } f(t) = u(t-3) \\
 &= (-1)^2 \frac{d^2}{ds^2} \mathcal{L}(f(t)) \quad (\because \text{multiplication by } t^2) \\
 &= \frac{d^2}{ds^2} \mathcal{L}(u(t-3)) \\
 &= \frac{d}{ds} \left(\frac{d}{ds} \left(\frac{e^{-3s}}{s} \right) \right) \\
 &= \frac{d}{ds} \left(\frac{s(e^{-3s})(-3) - e^{-3s}(1)}{s^2} \right) \quad \left(\because d\left(\frac{u}{v}\right) = \frac{v du - u dv}{v^2} \right) \\
 &= \frac{d}{ds} \left(-\frac{3e^{-3s}}{s} - \frac{e^{-3s}}{s^2} \right) \\
 &= \frac{d}{ds} \left(-\frac{3e^{-3s}}{s} \right) - \frac{d}{ds} \left(\frac{e^{-3s}}{s^2} \right) \\
 &= -3 \left(\frac{s(e^{-3s})(-3) - e^{-3s}(1)}{s^2} \right) - \left(\frac{s^2(e^{-3s})(-3) - e^{-3s}(2s)}{s^4} \right) \\
 &= -3 \left(-\frac{3e^{-3s}}{s} - \frac{e^{-3s}}{s^2} \right) - \left(-\frac{3e^{-3s}}{s^2} - \frac{2e^{-3s}}{s^3} \right) \\
 &= \frac{9e^{-3s}}{s} + \frac{3e^{-3s}}{s^2} + \frac{3e^{-3s}}{s^2} + \frac{2e^{-3s}}{s^3} \\
 &= \frac{1}{s^3} (2e^{-3s}) + \frac{1}{s^2} (6e^{-3s}) + \frac{1}{s} (9e^{-3s})
 \end{aligned}$$

$$\mathcal{L}(t^2 u(t-3)) = e^{-3s} \left(\frac{9}{s} + \frac{6}{s^2} + \frac{2}{s^3} \right)$$

9 Evaluation of Integrals using the Laplace Transform

Problems: Solve by L.T.

$$(1) \int_0^\infty t e^{-3t} \sin t \, dt \quad (2) \int_0^\infty \left(\frac{e^{-t} - e^{-3t}}{t} \right) dt$$

Solution (1): Given

$$I = \int_0^\infty t e^{-3t} \sin t \, dt$$

It is of the form

$$= \int_0^\infty e^{-st} f(t) \, dt \quad \text{where } s = 3, \quad f(t) = t \sin t$$

$$\begin{aligned}
 &= \left(\mathcal{L}(f(t)) \right)_{s=3} \\
 &= \left\{ \mathcal{L}(t \sin t) \right\}_{s=3}
 \end{aligned}$$

$$\begin{aligned}
&= \left\{ -\frac{d}{ds} \mathcal{L}(\sin t) \right\}_{s=3} \quad (\because \text{multiplication by } t) \\
&= \left\{ -\frac{d}{ds} \left(\frac{1}{s^2 + 1} \right) \right\}_{s=3} \\
&= - \left\{ (-1)(s^2 + 1)^{-2}(2s) \right\}_{s=3} \\
&= \left\{ \frac{2s}{(s^2 + 1)^2} \right\}_{s=3} \\
&= \frac{6}{(10)^2} \\
&= \frac{6}{100}
\end{aligned}$$

$$I = \int_0^\infty t e^{-3t} \sin t \, dt = \frac{3}{50}$$

Solution (2): Given

$$I = \int_0^\infty \left(\frac{e^{-t} - e^{-3t}}{t} \right) dt$$

It is of the form

$$\begin{aligned}
&= \int_0^\infty e^{-st} f(t) \, dt \quad \text{where } s = 0, \quad f(t) = \frac{e^{-t} - e^{-3t}}{t} \\
&= \left(\mathcal{L}(f(t)) \right)_{s=0} \\
&= \left\{ \mathcal{L} \left(\frac{e^{-t} - e^{-3t}}{t} \right) \right\}_{s=0} \\
&= \left\{ \int_s^\infty \mathcal{L}(e^{-t} - e^{-3t}) \, ds \right\}_{s=0} \quad (\because \text{division by } t) \\
&= \left\{ \int_s^\infty \left(\frac{1}{s+1} - \frac{1}{s+3} \right) ds \right\}_{s=0} \\
&= \left\{ [\log(s+1) - \log(s+3)]_s^\infty \right\}_{s=0} \\
&= \left\{ \left[\log \left(\frac{s+1}{s+3} \right) \right]_s^\infty \right\}_{s=0} \\
&= \left\{ \log(1) - \log \left(\frac{s+1}{s+3} \right) \right\}_{s=0} \\
&= \left\{ \log \left(\frac{s+3}{s+1} \right) \right\}_{s=0} \\
&= \log \left(\frac{3}{1} \right)
\end{aligned}$$

$$I = \int_0^\infty \left(\frac{e^{-t} - e^{-3t}}{t} \right) dt = \log 3$$

10 Inverse Laplace Transform

Inverse Laplace Transform

We know that $\mathcal{L}(f(t)) = F(s)$; hence the inverse Laplace transform is defined by

$$\mathcal{L}^{-1}(F(s)) = f(t)$$

Hence we have the following **standard inverse Laplace transforms**:

Standard Inverse Laplace Transforms

$$(1) \quad \mathcal{L}(1) = \frac{1}{s} \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{s}\right) = 1$$

$$(2) \quad \mathcal{L}(t^n) = \frac{n!}{s^{n+1}} \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{s^n}\right) = \frac{t^{n-1}}{(n-1)!}; \quad \text{in particular}$$

$$\mathcal{L}^{-1}\left(\frac{1}{s^2}\right) = t, \quad \mathcal{L}^{-1}\left(\frac{1}{s^3}\right) = \frac{t^2}{2!}, \quad \mathcal{L}^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{n!}$$

$$(3) \quad \mathcal{L}(e^{at}) = \frac{1}{s-a} \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{s-a}\right) = e^{at}$$

$$(4) \quad \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2} \Rightarrow \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2}\right) = \frac{1}{a} \sin at$$

$$(5) \quad \mathcal{L}(\cos at) = \frac{s}{s^2 + a^2} \Rightarrow \mathcal{L}^{-1}\left(\frac{s}{s^2 + a^2}\right) = \cos at$$

$$(6) \quad \mathcal{L}(u_a(t)) = \frac{e^{-as}}{s} \Rightarrow \mathcal{L}^{-1}\left(\frac{e^{-as}}{s}\right) = u_a(t)$$

$$(7) \quad \mathcal{L}(\delta_a(t)) = e^{-as} \Rightarrow \mathcal{L}^{-1}(e^{-as}) = \delta_a(t)$$

$$(8) \quad \mathcal{L}(\delta(t)) = 1 \Rightarrow \mathcal{L}^{-1}(1) = \delta(t)$$

11 Results on the Inverse Laplace Transform

Result (1): Inverse L.T. by first shifting property

We know that, by the first shifting property,

$$\mathcal{L}(e^{at}f(t)) = F(s-a)$$

$$\begin{aligned} \Rightarrow \mathcal{L}^{-1}(F(s-a)) &= e^{at}f(t) \\ &= e^{at} \mathcal{L}^{-1}(F(s)) \end{aligned}$$

$$\mathcal{L}^{-1}(F(s-a)) = e^{at} \mathcal{L}^{-1}(F(s)) \quad (\text{R1})$$

$$\Rightarrow \mathcal{L}^{-1}(F(s+a)) = e^{-at} \mathcal{L}^{-1}(F(s))$$

Result (2): Inverse L.T. by L.T. of derivatives (multiplication by s)

We know that, by the L.T. of derivatives,

$$\mathcal{L}(f'(t)) = sF(s) - f(0)$$

Assume $f(0) = 0$; then

$$\mathcal{L}^{-1}(s F(s)) = f'(t) = \frac{d}{dt} f(t)$$

$$\mathcal{L}^{-1}(s F(s)) = \frac{d}{dt} \mathcal{L}^{-1}(F(s)), \quad \text{provided } f(0) = 0 \quad (\text{R2})$$

This is known as *multiplication by s* . Similarly,

$$\mathcal{L}^{-1}(s^2 F(s)) = \frac{d^2}{dt^2} \mathcal{L}^{-1}(F(s)), \quad \text{provided } f(0) = 0, f'(0) = 0$$

and so on.

Result (3): Inverse L.T. by L.T. of integrals (division by s)

$$\mathcal{L}\left(\int_0^t f(t) dt\right) = \frac{F(s)}{s} \Rightarrow \mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t f(t) dt$$

$$\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t \mathcal{L}^{-1}(F(s)) dt \quad (\text{R3})$$

$$\text{Similarly } \mathcal{L}^{-1}\left(\frac{F(s)}{s^2}\right) = \int_0^t \int_0^t \mathcal{L}^{-1}(F(s)) dt dt$$

and so on (known as *division by s*).

Problems: Find $\mathcal{L}^{-1}$ of the following

$$(1) \frac{s}{(s+a)^2 + b^2} \quad (2) \frac{s^2}{s^2 + a^2} \quad (3) \frac{s^2}{(s-1)^4} \quad (4) \frac{1}{s(s^2 - 2s + 5)}$$

Solution (1):

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{s}{(s+a)^2 + b^2}\right) &= \mathcal{L}^{-1}\left(\frac{s+a-a}{(s+a)^2 + b^2}\right) \\ &= \mathcal{L}^{-1}\left(\frac{s+a}{(s+a)^2 + b^2}\right) - \mathcal{L}^{-1}\left(\frac{a}{(s+a)^2 + b^2}\right) \\ &= e^{-at} \mathcal{L}^{-1}\left(\frac{s}{s^2 + b^2}\right) - e^{-at} \mathcal{L}^{-1}\left(\frac{a}{s^2 + b^2}\right) \quad (\because \text{by Result (1)}) \\ &= e^{-at} \cos bt - e^{-at} a \cdot \frac{1}{b} \sin bt \end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{s}{(s+a)^2 + b^2}\right) = e^{-at} \left(\cos bt - \frac{a}{b} \sin bt \right)$$

Solution (2):

$$\begin{aligned}
\mathcal{L}^{-1}\left(\frac{s^2}{s^2+a^2}\right) &= \mathcal{L}^{-1}\left(s \cdot \frac{s}{s^2+a^2}\right) \\
&= \mathcal{L}^{-1}(s F(s)) \\
&= \frac{d}{dt} \mathcal{L}^{-1}(F(s)) \quad (\because \text{by Result (2)}) \\
&= \frac{d}{dt} \mathcal{L}^{-1}\left(\frac{s}{s^2+a^2}\right) = \frac{d}{dt}(\cos at)
\end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{s^2}{s^2+a^2}\right) = -a \sin at$$

Solution (3):

$$\begin{aligned}
\mathcal{L}^{-1}\left(\frac{s^2}{(s-1)^4}\right) &= \mathcal{L}^{-1}\left(s^2 \cdot \frac{1}{(s-1)^4}\right) \\
&= \frac{d^2}{dt^2} \mathcal{L}^{-1}\left(\frac{1}{(s-1)^4}\right) \quad (\because \text{by Result (2)}) \\
&= \frac{d^2}{dt^2} \left\{ e^t \mathcal{L}^{-1}\left(\frac{1}{s^4}\right) \right\} \quad (\text{by Result (1)}) \\
&= \frac{d^2}{dt^2} \left\{ e^t \frac{t^3}{3!} \right\} \quad \left(\because \mathcal{L}^{-1}\left(\frac{1}{s^n}\right) = \frac{t^{n-1}}{(n-1)!} \right) \\
&= \frac{1}{6} \frac{d}{dt} \left(\frac{d}{dt} (t^3 e^t) \right) \\
&= \frac{1}{6} \frac{d}{dt} (3t^2 e^t + t^3 e^t) \\
&= \frac{1}{6} \{ 6t e^t + 3t^2 e^t + 3t^2 e^t + t^3 e^t \}
\end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{s^2}{(s-1)^4}\right) = \frac{1}{6} e^t \{ t^3 + 6t^2 + 6t \}$$

Solution (4):

$$\begin{aligned}
\mathcal{L}^{-1}\left(\frac{1}{s(s^2-2s+5)}\right) &= \mathcal{L}^{-1}\left(\frac{1}{s} \cdot \frac{1}{s^2-2s+5}\right) \\
&= \mathcal{L}^{-1}\left(\frac{1}{s} \cdot F(s)\right) \\
&= \int_0^t \mathcal{L}^{-1}(F(s)) dt \quad (\because \text{by Result (3)}) \\
&= \int_0^t \mathcal{L}^{-1}\left(\frac{1}{s^2-2s+5}\right) dt \\
&= \int_0^t \mathcal{L}^{-1}\left(\frac{1}{(s-1)^2+4}\right) dt \\
&= \int_0^t e^t \mathcal{L}^{-1}\left(\frac{1}{s^2+2^2}\right) dt \quad (\text{by Result (1)})
\end{aligned}$$

$$\begin{aligned}
&= \int_0^t e^t \frac{1}{2} \sin 2t \, dt \\
&= \frac{1}{2} \int_0^t e^t \sin 2t \, dt \\
&= \frac{1}{2} \left[\frac{e^t}{1^2 + 2^2} (1 \cdot \sin 2t - 2 \cos 2t) \right]_0^t \quad \left(\because \int e^{at} \sin bt \, dt = \frac{e^{at}}{a^2 + b^2} (a \sin bt - b \cos bt) \right) \\
&= \frac{1}{2} \left[\frac{e^t}{5} (\sin 2t - 2 \cos 2t) + \frac{2}{5} \right]
\end{aligned}$$

$$\mathcal{L}^{-1} \left(\frac{1}{s(s^2 - 2s + 5)} \right) = \frac{e^t}{10} (\sin 2t - 2 \cos 2t) + \frac{1}{5}$$

Exercise: Find $\mathcal{L}^{-1}$ of the following

$$(1) \frac{s}{s^2 + 2s + 5} \quad (2) \frac{s^3}{s^2 + a^2} \quad (3) \frac{1}{s(s^2 + a^2)}$$

Result (4): Inverse L.T. by L.T. of multiplication by t

We know that, by the L.T. of multiplication by t ,

$$\begin{aligned}
\mathcal{L}(t f(t)) &= -\frac{d}{ds} F(s) = -F'(s) \\
\Rightarrow \mathcal{L}^{-1}(F'(s)) &= -t f(t)
\end{aligned}$$

$$\mathcal{L}^{-1}[F'(s)] = -t \mathcal{L}^{-1}[F(s)] \quad (\text{R4})$$

This result we can apply for the $F(s)$ where the differentiation of the denominator (Dr.) equals the numerator (Nr.). In such a case the given function of s will be treated as $F'(s)$.

Result (5)

From Result (4), we can write

$$\mathcal{L}^{-1}[F(s)] = -\frac{1}{t} \mathcal{L}^{-1}[F'(s)] \quad (\text{R5})$$

This result is applicable when the given $F(s)$ involves logarithmic or inverse trigonometric functions.

Problems: Find $\mathcal{L}^{-1}$ of the following

$$(1) \frac{s}{(s^2 + a^2)^2} \quad (2) \frac{s + 3}{(s^2 + 6s + 13)^2} \quad (3) \frac{2(s + 1)}{(s^2 + 2s + 2)^2}$$

Solution (1): Let

$$F'(s) = \frac{s}{(s^2 + a^2)^2} \Rightarrow \frac{dF(s)}{ds} = \frac{s}{(s^2 + a^2)^2} \Rightarrow dF(s) = \frac{s \, ds}{(s^2 + a^2)^2}$$

Integrating,

$$F(s) = \int \frac{s ds}{(s^2 + a^2)^2}$$

Let $u = s^2 + a^2 \Rightarrow du = 2s ds$

$$\Rightarrow F(s) = \int \frac{du/2}{u^2} = \frac{1}{2} \int u^{-2} du$$

$$F(s) = \frac{1}{2} \left(\frac{u^{-1}}{-1} \right) = -\frac{1}{2u}$$

$$\Rightarrow F(s) = -\frac{1}{2(s^2 + a^2)}$$

By Result (4) we get $\mathcal{L}^{-1}[F'(s)] = -t \mathcal{L}^{-1}[F(s)]$

$$\Rightarrow \mathcal{L}^{-1}\left[\frac{s}{(s^2 + a^2)^2}\right] = -t \mathcal{L}^{-1}\left[-\frac{1}{2(s^2 + a^2)}\right] = \frac{t}{2} \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2}\right)$$

$$= \frac{t}{2} \cdot \frac{1}{a} \sin at$$

$$\mathcal{L}^{-1}\left(\frac{s}{(s^2 + a^2)^2}\right) = \frac{t}{2a} \sin at$$

Solution (2): Let

$$F'(s) = \frac{s+3}{(s^2 + 6s + 13)^2} \Rightarrow \frac{dF(s)}{ds} = \frac{s+3}{(s^2 + 6s + 13)^2}$$

$$F(s) = \int \frac{(s+3)}{(s^2 + 6s + 13)^2} ds$$

Let $u = s^2 + 6s + 13 \Rightarrow du = (2s + 6) ds = 2(s + 3) ds$

$$= \int \frac{du/2}{u^2} = \frac{1}{2} \int u^{-2} du = \frac{1}{2} \cdot \frac{u^{-1}}{-1} = -\frac{1}{2u}$$

$$F(s) = \frac{1}{-2(s^2 + 6s + 13)}$$

By Result (4),

$$\begin{aligned} \mathcal{L}^{-1}(F'(s)) &= -t \mathcal{L}^{-1}(F(s)) \\ &= -t \mathcal{L}^{-1}\left(\frac{1}{-2(s^2 + 6s + 13)}\right) \\ &= \frac{t}{2} \mathcal{L}^{-1}\left(\frac{1}{s^2 + 6s + 13}\right) \\ &= \frac{t}{2} \mathcal{L}^{-1}\left(\frac{1}{(s+3)^2 + 4}\right) \\ &= \frac{t}{2} e^{-3t} \mathcal{L}^{-1}\left(\frac{1}{s^2 + 2^2}\right) \quad (\text{by Result (1)}) \end{aligned}$$

$$\begin{aligned}
&= \frac{t}{2} e^{-3t} \frac{1}{2} \sin 2t \\
&= \frac{t}{4} e^{-3t} \sin 2t
\end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{s+3}{(s^2+6s+13)^2}\right) = \frac{t}{4} e^{-3t} \sin 2t$$

Solution (3): Proceeding in the same way,

$$\text{Ans: } \mathcal{L}^{-1}\left(\frac{2(s+1)}{(s^2+2s+2)^2}\right) = e^{-t} t \sin t$$

Problems: Find $\mathcal{L}^{-1}$ of the following

$$(1) \log_e\left(\frac{s+1}{s-1}\right) \quad (2) \tan^{-1}\left(\frac{s}{2}\right)$$

Solution (1): Let

$$\begin{aligned}
F(s) &= \log_e\left(\frac{s+1}{s-1}\right) \\
\Rightarrow F(s) &= \log(s+1) - \log(s-1) \\
\Rightarrow \frac{d}{ds}F(s) &= \frac{1}{s+1} - \frac{1}{s-1} = F'(s)
\end{aligned}$$

By Result (5), $\mathcal{L}^{-1}(F(s)) = -\frac{1}{t} \mathcal{L}^{-1}(F'(s))$

$$\begin{aligned}
\Rightarrow \mathcal{L}^{-1}\left(\log\left(\frac{s+1}{s-1}\right)\right) &= -\frac{1}{t} \mathcal{L}^{-1}\left(\frac{1}{s+1} - \frac{1}{s-1}\right) \\
&= -\frac{1}{t} \left\{ \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) - \mathcal{L}^{-1}\left(\frac{1}{s-1}\right) \right\} \\
&= -\frac{1}{t} \{e^{-t} - e^t\}
\end{aligned}$$

$$\mathcal{L}^{-1}\left(\log\left(\frac{s+1}{s-1}\right)\right) = \frac{e^t - e^{-t}}{t}$$

Solution (2): Let $F(s) = \tan^{-1}\left(\frac{s}{2}\right)$

$$\begin{aligned}
\frac{d}{ds}F(s) &= \frac{1}{1 + \left(\frac{s}{2}\right)^2} \cdot \left(\frac{1}{2}\right) \\
\Rightarrow F'(s) &= \frac{1}{2} \left(\frac{2^2}{2^2 + s^2}\right) \cdot \frac{1}{2} \cdot 2 = \frac{2}{s^2 + 2^2}
\end{aligned}$$

By Result (5), $\mathcal{L}^{-1}(F(s)) = -\frac{1}{t} \mathcal{L}^{-1}(F'(s))$

$$\Rightarrow \mathcal{L}^{-1}\left(\tan^{-1}\left(\frac{s}{2}\right)\right) = -\frac{1}{t} \mathcal{L}^{-1}\left(\frac{2}{s^2 + 2^2}\right)$$

$$= -\frac{1}{t} \sin 2t$$

$$\mathcal{L}^{-1}\left(\tan^{-1}\left(\frac{s}{2}\right)\right) = -\frac{\sin 2t}{t} //$$

Result (6): Inverse L.T. by the second shifting property

We know that, by the second shifting property,

$$\mathcal{L}(f(t-a)u(t-a)) = e^{-as}F(s)$$

$$\begin{aligned} \Rightarrow \mathcal{L}^{-1}(e^{-as}F(s)) &= f(t-a)u(t-a) \\ &= u(t-a)(f(t))_{t \rightarrow t-a} \end{aligned}$$

$$\mathcal{L}^{-1}(e^{-as}F(s)) = u(t-a) \left(\mathcal{L}^{-1}(F(s)) \right)_{t \rightarrow t-a} \quad (\text{R6})$$

Problems: Find $\mathcal{L}^{-1}$ of the following

$$(1) \frac{e^{-\pi s} s}{s^2 + 1} \quad (2) \frac{e^{-s}}{(s+1)^3}$$

Solution (1):

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{e^{-\pi s} s}{s^2 + 1}\right) &= u(t-\pi) \left[\mathcal{L}^{-1}\left(\frac{s}{s^2 + 1}\right) \right]_{t \rightarrow t-\pi} \quad (\because \text{by Result (6)}) \\ &= u(t-\pi) (\cos t)_{t \rightarrow t-\pi} \\ &= u(t-\pi) \cos(t-\pi) \end{aligned}$$

Solution (2):

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{e^{-s}}{(s+1)^3}\right) &= u(t-1) \left\{ \mathcal{L}^{-1}\left(\frac{1}{(s+1)^3}\right) \right\}_{t \rightarrow t-1} \quad (\because \text{by Result (6)}) \\ &= u(t-1) \left\{ e^{-t} \mathcal{L}^{-1}\left(\frac{1}{s^3}\right) \right\}_{t \rightarrow t-1} \\ &= u(t-1) \left\{ e^{-t} \frac{t^2}{2!} \right\}_{t \rightarrow t-1} \\ &= u(t-1) \left\{ \frac{e^{-(t-1)}(t-1)^2}{2} \right\} \\ &= \frac{1}{2} (t-1)^2 e^{-(t-1)} u(t-1) \end{aligned}$$

12 Convolution

Convolution of two functions

If $f(t)$ and $g(t)$ are two continuous functions then the convolution of $f(t)$ and $g(t)$ is defined by

$$f(t) * g(t) = \int_0^t f(u) g(t-u) du$$

Convolution Theorem

If $\mathcal{L}(f(t)) = F(s)$ and $\mathcal{L}(g(t)) = G(s)$ then

$$\mathcal{L}(f(t) * g(t)) = F(s) \cdot G(s)$$

By the above we get

$$\begin{aligned} \mathcal{L}^{-1}(F(s) \cdot G(s)) &= f(t) * g(t) \\ &= \mathcal{L}^{-1}(F(s)) * \mathcal{L}^{-1}(G(s)) \end{aligned}$$

$$\mathcal{L}^{-1}(F(s) \cdot G(s)) = \mathcal{L}^{-1}(F(s)) * \mathcal{L}^{-1}(G(s))$$

Problems: Find $\mathcal{L}^{-1}$ of the following by the Convolution Theorem

$$(1) \frac{s^2}{(s^2 + a^2)^2} \quad (2) \frac{1}{(s+1)(s+9)^2}$$

Solution (1):

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{s^2}{(s^2 + a^2)^2}\right) &= \mathcal{L}^{-1}\left(\frac{s}{(s^2 + a^2)} \cdot \frac{s}{(s^2 + a^2)}\right) \\ &= \mathcal{L}^{-1}(F(s) G(s)) \\ &= \mathcal{L}^{-1}(F(s)) * \mathcal{L}^{-1}(G(s)) \\ &= \mathcal{L}^{-1}\left(\frac{s}{s^2 + a^2}\right) * \mathcal{L}^{-1}\left(\frac{s}{s^2 + a^2}\right) \\ &= \cos at * \cos at \\ &= \int_0^t \cos au \cos a(t-u) du \quad \left(\because f(t) * g(t) = \int_0^t f(u) g(t-u) du\right) \\ &= \int_0^t \cos au \cos(at-au) du \\ &= \int_0^t \frac{1}{2} [\cos(au + at - au) + \cos(au - at + au)] du \\ &\quad \left(\because \cos A \cos B = \frac{1}{2} (\cos(A+B) + \cos(A-B))\right) \\ &= \frac{1}{2} \int_0^t (\cos at + \cos(2au - at)) du \\ &= \frac{1}{2} \left\{ \cos at (u)_0^t + \left(\frac{\sin(2au - at)}{2a}\right)_0^t \right\} \\ &= \frac{1}{2} \left\{ t \cos at + \frac{1}{2a} (\sin at - (-\sin at)) \right\} \end{aligned}$$

$$= \frac{1}{2} \left\{ t \cos at + \frac{1}{2a} (2 \sin at) \right\}$$

$$\mathcal{L}^{-1} \left(\frac{s^2}{(s^2 + a^2)^2} \right) = \frac{1}{2a} \left\{ at \cos at + \sin at \right\}$$

Solution (2):

$$\begin{aligned} \mathcal{L}^{-1} \left(\frac{1}{(s+1)(s+9)^2} \right) &= \mathcal{L}^{-1} \left(\frac{1}{s+1} \cdot \frac{1}{(s+9)^2} \right) \\ &= \mathcal{L}^{-1} \left(\frac{1}{s+1} \right) * \mathcal{L}^{-1} \left(\frac{1}{(s+9)^2} \right) \quad (\because \text{by Convolution Theorem}) \\ &= e^{-t} * e^{-9t} \mathcal{L}^{-1} \left(\frac{1}{s^2} \right) \\ &= e^{-t} * e^{-9t} \frac{t^1}{1!} \\ &= e^{-t} * e^{-9t} t \\ &= \int_0^t e^{-u} e^{-9(t-u)} (t-u) du \quad \left(\because f(t) * g(t) = \int_0^t f(u) g(t-u) du \right) \\ &= \int_0^t e^{-u} e^{-9t} e^{9u} (t-u) du \\ &= e^{-9t} \int_0^t e^{8u} (t-u) du \\ &= e^{-9t} \left\{ \int_0^t t e^{8u} du - \int_0^t u e^{8u} du \right\} \\ &= e^{-9t} \left\{ t \left(\frac{e^{8u}}{8} \right)_0^t - \left[u \frac{e^{8u}}{8} - 1 \cdot \frac{e^{8u}}{64} + 0 \right]_0^t \right\} \quad (\text{Bernoulli's formula}) \\ &= e^{-9t} \left\{ \frac{t}{8} (e^{8t} - 1) - \frac{1}{8} (t e^{8t} - 0) + \frac{1}{64} (e^{8t} - 1) \right\} \\ &= e^{-9t} \left\{ -\frac{t}{8} + \frac{1}{64} (e^{8t} - 1) \right\} \end{aligned}$$

$$\mathcal{L}^{-1} \left(\frac{1}{(s+1)(s+9)^2} \right) = \frac{e^{-9t}}{64} \left\{ e^{8t} - 8t - 1 \right\}$$

13 Partial Fraction Method for the Inverse L.T.

Problems

$$(1) \frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \quad (2) \frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}$$

Solution (1): Given

$$\frac{s^2}{(s^2 + a^2)(s^2 + b^2)} = \frac{A}{s^2 + a^2} + \frac{B}{s^2 + b^2} = \frac{A(s^2 + b^2) + B(s^2 + a^2)}{(s^2 + a^2)(s^2 + b^2)}$$

$$\Rightarrow s^2 = A(s^2 + b^2) + B(s^2 + a^2)$$

Put $s^2 = -a^2$:

$$\Rightarrow -a^2 = A(-a^2 + b^2) \Rightarrow \boxed{A = \frac{a^2}{a^2 - b^2}}$$

Put $s^2 = -b^2$:

$$-b^2 = B(-b^2 + a^2) \Rightarrow \boxed{B = \frac{-b^2}{a^2 - b^2}}$$

$$\Rightarrow \frac{s^2}{(s^2 + a^2)(s^2 + b^2)} = \frac{a^2}{a^2 - b^2} \left(\frac{1}{s^2 + a^2} \right) - \frac{b^2}{a^2 - b^2} \left(\frac{1}{s^2 + b^2} \right)$$

$$\Rightarrow \mathcal{L}^{-1} \left(\frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right) = \frac{a^2}{a^2 - b^2} \mathcal{L}^{-1} \left(\frac{1}{s^2 + a^2} \right) - \frac{b^2}{a^2 - b^2} \mathcal{L}^{-1} \left(\frac{1}{s^2 + b^2} \right)$$

$$= \frac{a^2}{a^2 - b^2} \left(\frac{1}{a} \sin at \right) - \frac{b^2}{a^2 - b^2} \left(\frac{1}{b} \sin bt \right)$$

$$\boxed{\mathcal{L}^{-1} \left(\frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right) = \frac{1}{a^2 - b^2} (a \sin at - b \sin bt)}$$

Solution (2):

$$\frac{5s^2 - 15s - 11}{(s + 1)(s - 2)^3} = \frac{A}{s + 1} + \frac{B}{s - 2} + \frac{C}{(s - 2)^2} + \frac{D}{(s - 2)^3}$$

$$\frac{5s^2 - 15s - 11}{(s + 1)(s - 2)^3} = \frac{A(s - 2)^3 + B(s + 1)(s - 2)^2 + C(s + 1)(s - 2) + D(s + 1)}{(s + 1)(s - 2)^3}$$

$$\Rightarrow 5s^2 - 15s - 11 = A(s - 2)^3 + B(s + 1)(s - 2)^2 + C(s + 1)(s - 2) + D(s + 1)$$

Put $s = -1$:

$$5 + 15 - 11 = A(-1 - 2)^3 \Rightarrow 9 = A(-27) \Rightarrow \boxed{A = -\frac{1}{3}}$$

Put $s = 2$:

$$20 - 30 - 11 = D(3) \Rightarrow -21 = 3D \Rightarrow \boxed{D = -7}$$

Compare s^3 :

$$0 = A + B \Rightarrow B = -A = \frac{1}{3} \Rightarrow \boxed{B = \frac{1}{3}}$$

Compare s^2 :

$$\boxed{C = 4}$$

Therefore

$$\mathcal{L}^{-1} \left(\frac{5s^2 - 15s - 11}{(s + 1)(s - 2)^3} \right) = -\frac{1}{3} \mathcal{L}^{-1} \left(\frac{1}{s + 1} \right) + \frac{1}{3} \mathcal{L}^{-1} \left(\frac{1}{s - 2} \right) + 4 \mathcal{L}^{-1} \left(\frac{1}{(s - 2)^2} \right) - 7 \mathcal{L}^{-1} \left(\frac{1}{(s - 2)^3} \right)$$

$$= -\frac{1}{3} e^{-t} + \frac{1}{3} e^{2t} + 4 e^{2t} t - 7 e^{2t} \frac{t^2}{2}$$

14 Solution of ODEs with Boundary Conditions

Recall: By the L.T. of derivatives, we know that

$$\begin{aligned}\mathcal{L}(f'(t)) &= sF(s) - f(0) \\ \mathcal{L}(f''(t)) &= s^2F(s) - sf(0) - f'(0) \\ \mathcal{L}(f'''(t)) &= s^3F(s) - s^2f(0) - sf'(0) - f''(0)\end{aligned}$$

and so on.

Problem 1

Solve $y''' - 3y'' + 3y' - y = t^2e^t$, given $y(0) = 1$, $y'(0) = 0$, $y''(0) = -2$.

Solution: Apply the L.T. on the given differential equation:

$$\mathcal{L}(y''') - 3\mathcal{L}(y'') + 3\mathcal{L}(y') - \mathcal{L}(y) = \mathcal{L}(t^2e^t)$$

$$\begin{aligned}\Rightarrow \left[s^3Y(s) - s^2y(0) - sy'(0) - y''(0) \right] - 3 \left[s^2Y(s) - sy(0) - y'(0) \right] \\ + 3 \left[sY(s) - y(0) \right] - Y(s) = \left[\mathcal{L}(t^2) \right]_{s \rightarrow s-1}\end{aligned}$$

Now apply the boundary conditions:

$$\Rightarrow \left(s^3Y(s) - s^2(1) - s(0) + 2 \right) - 3 \left(s^2Y(s) - s - 0 \right) + 3 \left(sY(s) - 1 \right) - Y(s) = \left(\frac{2}{s^3} \right)_{s \rightarrow s-1}$$

$$\begin{aligned}\Rightarrow Y(s)(s^3 - 3s^2 + 3s - 1) + (-s^2 + 2 + 3s - 3) &= \frac{2}{(s-1)^3} \\ \Rightarrow Y(s)(s-1)^3 - (s^2 - 3s + 1) &= \frac{2}{(s-1)^3} \\ \Rightarrow Y(s)(s-1)^3 &= \frac{2}{(s-1)^3} + (s^2 - 3s + 1) \\ \Rightarrow Y(s) &= \frac{2}{(s-1)^6} + \frac{s^2 - 3s + 1}{(s-1)^3} \\ \Rightarrow \mathcal{L}(y(t)) &= \frac{2}{(s-1)^6} + \frac{s^2 - 3s + 1}{(s-1)^3} \\ \therefore y(t) &= \mathcal{L}^{-1} \left[\frac{2}{(s-1)^6} + \frac{s^2 - 3s + 1}{(s-1)^3} \right] \quad (1)\end{aligned}$$

Now resolve into partial fractions:

$$\frac{s^2 - 3s + 1}{(s-1)^3} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{(s-1)^3} = \frac{A(s-1)^2 + B(s-1) + C}{(s-1)^3}$$

$$\Rightarrow s^2 - 3s + 1 = A(s-1)^2 + B(s-1) + C$$

Put $s = 1$:

$$1 - 3 + 1 = C \Rightarrow \boxed{C = -1}$$

Compare s^2 : $\boxed{1 = A}$

Compare s : $-3 = -2A + B$

$$\Rightarrow B = -3 + 2(1) \quad (\because A = 1) \quad \boxed{B = -1}$$

$$\therefore \frac{s^2 - 3s + 1}{(s-1)^3} = \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} \quad \text{Sub. in (1)}$$

$$\begin{aligned} \Rightarrow y(t) &= \mathcal{L}^{-1}\left(\frac{2}{(s-1)^6}\right) + \mathcal{L}^{-1}\left(\frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3}\right) \\ &= 2e^t \mathcal{L}^{-1}\left(\frac{1}{s^6}\right) + e^t - e^t \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) - e^t \mathcal{L}^{-1}\left(\frac{1}{s^3}\right) \\ &= 2e^t \frac{t^5}{5!} + e^t - e^t t - e^t \frac{t^2}{2!} \quad \left(\because \mathcal{L}^{-1}\left(\frac{1}{s^n}\right) = \frac{t^{n-1}}{(n-1)!}\right) \\ &= e^t \left(\frac{2t^5}{120} + 1 - t - \frac{t^2}{2}\right) \end{aligned}$$

$$y(t) = e^t \left(1 - t - \frac{t^2}{2} + \frac{t^5}{60}\right)$$

Problem 2

Solve $y^{(iv)} - y = 0$, given $y(0) = y'(0) = y''(0) = 0$ and $y'''(0) = 2$.

Solution: Apply the L.T. on the given differential equation:

$$\mathcal{L}(y^{(iv)}) - \mathcal{L}(y) = 0$$

$$\Rightarrow [s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0)] - Y(s) = 0$$

Apply the boundary conditions; we get

$$\begin{aligned} (s^4 Y(s) - 2) - Y(s) &= 0 \\ (s^4 - 1)Y(s) &= 2 \\ \Rightarrow Y(s) &= \frac{2}{s^4 - 1} \\ \Rightarrow \mathcal{L}(y(t)) &= \frac{2}{s^4 - 1} = \frac{2}{(s^2 + 1)(s^2 - 1)} \end{aligned} \quad (1)$$

Now resolve into partial fractions:

$$\frac{2}{(s^2 + 1)(s^2 - 1)} = \frac{A}{s^2 + 1} + \frac{B}{s^2 - 1}$$

Put $s^2 = -1$ and $s^2 = 1$:

$$(A = -1, B = 1) \quad \Rightarrow \quad \frac{2}{(s^2 + 1)(s^2 - 1)} = \frac{-1}{s^2 + 1} + \frac{1}{s^2 - 1}$$

$$\begin{aligned} \therefore (1) \Rightarrow \mathcal{L}(y(t)) &= \left[\frac{1}{s^2 - 1} - \frac{1}{s^2 + 1} \right] \\ \Rightarrow y(t) &= \mathcal{L}^{-1}\left(\frac{1}{s^2 - 1}\right) - \mathcal{L}^{-1}\left(\frac{1}{s^2 + 1}\right) \\ &= \sinh t - \sin t \end{aligned}$$

$$y(t) = \sinh t - \sin t$$

Exercise: Problem 3

Solve $y'' + 4y = \sin \omega t$, ω is constant, given that $y(0) = 0$ and $y'(0) = 0$.

15 Solution of Simultaneous ODEs**Problem 1**

Solve

$$\left. \begin{array}{l} x'' - y' = \cos t \\ x' + y'' = -\sin t \end{array} \right\} \quad \text{given that } x(0) = 1, y(0) = 0, x'(0) = 0, y'(0) = 1.$$

Solution: Apply the L.T. on the given set of ODEs:

$$\Rightarrow \begin{cases} \mathcal{L}(x'') - \mathcal{L}(y') = \mathcal{L}(\cos t) \\ \mathcal{L}(x') + \mathcal{L}(y'') = -\mathcal{L}(\sin t) \end{cases}$$

$$\Rightarrow \begin{cases} [s^2 X(s) - s x(0) - x'(0)] - [s Y(s) - y(0)] = \frac{s}{s^2 + 1} \\ [s X(s) - x(0)] + [s^2 Y(s) - s y(0) - y'(0)] = -\frac{1}{s^2 + 1} \end{cases}$$

Apply the boundary conditions:

$$\begin{cases} (s^2 X(s) - s) - s Y(s) = \frac{s}{s^2 + 1} \\ (s X(s) - 1) + (s^2 Y(s) - 1) = -\frac{1}{s^2 + 1} \end{cases}$$

$$\Rightarrow \begin{cases} s^2 X(s) - s Y(s) = \frac{s}{s^2 + 1} + s & \text{--- (1)} \\ s X(s) + s^2 Y(s) = -\frac{1}{s^2 + 1} + 2 & \text{--- (2)} \end{cases}$$

(1) $\times s$:

$$s^3 X(s) - s^2 Y(s) = \frac{s^2}{s^2 + 1} + s^2$$

(2):

$$s X(s) + s^2 Y(s) = -\frac{1}{s^2 + 1} + 2$$

Adding,

$$\begin{aligned} \Rightarrow (s^3 + s)X(s) &= \frac{s^2}{s^2 + 1} - \frac{1}{s^2 + 1} + (s^2 + 2) \\ \Rightarrow s(s^2 + 1)X(s) &= \frac{s^2}{s^2 + 1} - \frac{1}{s^2 + 1} + (s^2 + 2) \\ X(s) &= \frac{s}{(s^2 + 1)^2} - \frac{1}{s(s^2 + 1)^2} + \frac{s^2 + 2}{s(s^2 + 1)} \\ \mathcal{L}(x(t)) &= \frac{s}{(s^2 + 1)^2} - \frac{1}{s(s^2 + 1)^2} + \frac{s}{s^2 + 1} + \frac{2}{s(s^2 + 1)} \\ \Rightarrow x(t) &= \mathcal{L}^{-1}\left(\frac{s}{(s^2 + 1)^2}\right) - \mathcal{L}^{-1}\left(\frac{1}{s(s^2 + 1)^2}\right) + \mathcal{L}^{-1}\left(\frac{s}{s^2 + 1}\right) + 2\mathcal{L}^{-1}\left(\frac{1}{s(s^2 + 1)}\right) \end{aligned}$$

$$= \frac{t}{2} \sin t - \mathcal{L}^{-1}\left(\frac{1}{s(s^2+1)^2}\right) + \cos t + 2\mathcal{L}^{-1}\left(\frac{1}{s(s^2+1)}\right) \quad (3)$$

$$\left(\because \mathcal{L}^{-1}\left(\frac{s}{(s^2+a^2)^2}\right) = \frac{t}{2a} \sin at\right)$$

Now

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{1}{s(s^2+1)}\right) &= \mathcal{L}^{-1}\left(\frac{1}{s} F(s)\right) \quad \text{where } F(s) = \frac{1}{s^2+1} \\ &= \int_0^t \mathcal{L}^{-1}(F(s)) dt \\ &= \int_0^t \mathcal{L}^{-1}\left(\frac{1}{s^2+1}\right) dt = \int_0^t \sin t dt \\ &= (-\cos t)_0^t = -(\cos t - 1) \end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{1}{s(s^2+1)}\right) = 1 - \cos t \quad (4)$$

Next,

$$\begin{aligned} \mathcal{L}^{-1}\left(\frac{1}{s(s^2+1)^2}\right) &= \mathcal{L}^{-1}\left(\frac{s}{s^2(s^2+1)^2}\right) \\ &= \mathcal{L}^{-1}\left(\frac{1}{s^2} F(s)\right) \quad \text{where } F(s) = \frac{s}{(s^2+1)^2} \\ &= \int_0^t \int_0^t \mathcal{L}^{-1}(F(s)) dt dt \\ &= \int_0^t \int_0^t \mathcal{L}^{-1}\left(\frac{s}{(s^2+1)^2}\right) dt dt \\ &= \int_0^t \int_0^t \left(\frac{t}{2} \sin t\right) dt dt \quad \left(\because \mathcal{L}^{-1}\left(\frac{s}{(s^2+a^2)^2}\right) = \frac{t}{2a} \sin at\right) \\ &= \frac{1}{2} \int_0^t \left(t(-\cos t) - 1(-\sin t) + 0\right)_0^t dt \quad (uv_1 - u'v_2 + u''v_3 - \dots) \\ &= \frac{1}{2} \int_0^t \left(-(t \cos t)_0^t + (\sin t)_0^t\right) dt \\ &= \frac{1}{2} \int_0^t \left(-(t \cos t - 0) + (\sin t - 0)\right) dt \\ &= \frac{1}{2} \int_0^t (\sin t - t \cos t) dt \\ &= \frac{1}{2} \left\{ (-\cos t)_0^t - [t(\sin t) - 1(-\cos t) + 0]_0^t \right\} \\ &= \frac{1}{2} \left\{ -(\cos t - 1) - [(t \sin t - 0) + (\cos t - 1)] \right\} \\ &= \frac{1}{2} \left\{ 1 - \cos t - t \sin t + 1 - \cos t \right\} \\ &= \frac{1}{2} \left\{ 2 - 2 \cos t - t \sin t \right\} \end{aligned}$$

$$\mathcal{L}^{-1}\left(\frac{1}{s(s^2 + 1)^2}\right) = 1 - \cos t - \frac{t}{2} \sin t \quad (5)$$

Substitute (4), (5) in (3):

$$\begin{aligned} x(t) &= \frac{t}{2} \sin t - 1 + \cos t + \frac{t}{2} \sin t + \cos t + 2(1 - \cos t) \\ &= t \sin t + 1 \end{aligned}$$

$$x(t) = 1 + t \sin t$$

Likewise,

$$y(t) = t \cos t$$

Exercise: Problem 2

Solve

$$\left. \begin{aligned} 2x + y' &= \sin 2t \\ x' - 2y &= \cos 2t \end{aligned} \right\} \quad \text{given } x(0) = 1 \text{ and } y(0) = 0.$$

16 Application to Electric Circuits

Current $I(t)$ at any time t :

Governing equations

- (1) **RL-circuit:** $\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L}$
- (2) **RC-circuit:** $\frac{dQ}{dt} + \frac{1}{RC} Q = \frac{E(t)}{R}, \quad \text{and } I = \frac{dQ}{dt}$
- (3) **LCR-circuit:** $L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = E(t)$

Problem 1

A resistance R in series with inductance L is connected with e.m.f. E . If the switch is connected at $t = 0$ and disconnected at $t = a$, find the current I in terms of t .

Solution: Since the circuit contains R and L , the current $I(t)$ is given by

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L} \quad (1)$$

Since $E(t) = E$ and also based on the given conditions, the flow of the voltage in the circuit is given by

$$E(t) = \begin{cases} E, & 0 < t < a \\ 0, & t > a \end{cases}$$

Apply the L.T. on (1):

$$\begin{aligned}
 \mathcal{L}(I'(t)) + \frac{R}{L} \mathcal{L}(I) &= \frac{1}{L} \mathcal{L}(E(t)) \\
 \Rightarrow [sI(s) - I(0)] + \frac{R}{L} I(s) &= \frac{1}{L} \left\{ \int_0^a E e^{-st} dt + \int_a^\infty 0 dt \right\} \\
 \Rightarrow sI(s) + \frac{R}{L} I(s) &= \frac{1}{L} \left\{ E \left(\frac{e^{-st}}{-s} \right)_0^a \right\} \quad (\because I(0) = 0) \\
 \Rightarrow \left(s + \frac{R}{L} \right) I(s) &= -\frac{1}{sL} E (e^{-as} - 1) \\
 \Rightarrow \left(s + \frac{R}{L} \right) I(s) &= +\frac{E}{L} \left(\frac{1 - e^{-as}}{s} \right) \\
 \Rightarrow I(s) &= \frac{E}{L} \left[\frac{1 - e^{-as}}{s(s + \frac{R}{L})} \right] \\
 \Rightarrow \mathcal{L}(I(t)) &= \frac{E}{L} \left(\frac{1 - e^{-as}}{s(s + \frac{R}{L})} \right) \tag{2}
 \end{aligned}$$

Now

$$\frac{1}{s(s + \frac{R}{L})} = \frac{A}{s} + \frac{B}{s + \frac{R}{L}} \Rightarrow 1 = A\left(s + \frac{R}{L}\right) + Bs$$

Put $s = 0$:

$$1 = A \frac{R}{L} \Rightarrow \boxed{A = \frac{L}{R}}$$

Put $s = -\frac{R}{L}$: then

$$1 = B\left(-\frac{R}{L}\right) \Rightarrow \boxed{B = -\frac{L}{R}}$$

$$\therefore \frac{1}{s(s + \frac{R}{L})} = \frac{L/R}{s} - \frac{L/R}{s + \frac{R}{L}} = \frac{L}{R} \left(\frac{1}{s} - \frac{1}{s + \frac{R}{L}} \right) \quad \text{Sub. in (2)}$$

$$\begin{aligned}
 \mathcal{L}(I(t)) &= \frac{E}{L} (1 - e^{-as}) \left[\frac{L}{R} \left(\frac{1}{s} - \frac{1}{s + \frac{R}{L}} \right) \right] \\
 &= \frac{E}{R} \left\{ \frac{1 - e^{-as}}{s} - \frac{1 - e^{-as}}{s + \frac{R}{L}} \right\} \\
 \mathcal{L}(I(t)) &= \frac{E}{R} \left\{ \frac{1}{s} - \frac{e^{-as}}{s} - \frac{1}{s + \frac{R}{L}} + \frac{e^{-as}}{s + \frac{R}{L}} \right\} \\
 \Rightarrow I(t) &= \frac{E}{R} \left[\mathcal{L}^{-1}\left(\frac{1}{s}\right) - \mathcal{L}^{-1}\left(\frac{e^{-as}}{s}\right) - \mathcal{L}^{-1}\left(\frac{1}{s + \frac{R}{L}}\right) + \mathcal{L}^{-1}\left(\frac{e^{-as}}{s + \frac{R}{L}}\right) \right] \\
 &= \frac{E}{R} \left[1 - u(t - a) - e^{-\frac{R}{L}t} + \left(\mathcal{L}^{-1}\left(\frac{1}{s + \frac{R}{L}}\right) \right)_{t \rightarrow t-a} u(t - a) \right] \\
 &\quad \left(\because \mathcal{L}^{-1}(e^{-as} F(s)) = \left(\mathcal{L}^{-1}(F(s)) \right)_{t \rightarrow t-a} u(t - a) \right)
 \end{aligned}$$

$$\begin{aligned}
&= \frac{E}{R} \left[1 - u(t-a) - e^{-\frac{R}{L}t} + \left(e^{-\frac{R}{L}t} \right)_{t \rightarrow t-a} u(t-a) \right] \\
&= \frac{E}{R} \left[1 - u(t-a) - e^{-\frac{R}{L}t} + u(t-a) e^{-\frac{R}{L}(t-a)} \right]
\end{aligned}$$

$$I(t) = \frac{E}{R} \left[\left(1 - e^{-\frac{R}{L}t} \right) - u(t-a) \left(1 - e^{-\frac{R}{L}(t-a)} \right) \right]$$

Problem 2

A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show that the current at time t is $\frac{E}{R-aL} \left(e^{-at} - e^{-\frac{R}{L}t} \right)$.

Solution: Given $E(t) = Ee^{-at}$ and the circuit is an RL circuit; hence its differential equation is

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L} \Rightarrow \frac{dI}{dt} + \frac{R}{L} I = \frac{Ee^{-at}}{L} \quad (1)$$

Apply the L.T. on (1):

$$\begin{aligned}
sI(s) - I(0) + \frac{R}{L} I(s) &= \frac{E}{L} \cdot \frac{1}{s+a} \\
\Rightarrow \left(s + \frac{R}{L} \right) I(s) &= \frac{E}{L} \frac{1}{s+a} \quad (\because I(0) = 0) \\
\Rightarrow I(s) &= \frac{E}{L} \frac{1}{(s+a)(s+\frac{R}{L})} \quad (2)
\end{aligned}$$

Now

$$\frac{1}{(s+a)(s+\frac{R}{L})} = \frac{A}{s+a} + \frac{B}{s+\frac{R}{L}}, \quad A = \frac{1}{\frac{R}{L}-a}, \quad B = \frac{-1}{\frac{R}{L}-a}$$

$$\begin{aligned}
\therefore (2) \Rightarrow I(s) &= \frac{E}{L} \left[\frac{L}{R-aL} \cdot \frac{1}{s+a} - \frac{L}{R-aL} \cdot \frac{1}{s+\frac{R}{L}} \right] \\
\Rightarrow I(t) &= \frac{E}{R-aL} \left[\mathcal{L}^{-1} \left(\frac{1}{s+a} \right) - \mathcal{L}^{-1} \left(\frac{1}{s+\frac{R}{L}} \right) \right]
\end{aligned}$$

$$I(t) = \frac{E}{R-aL} \left[e^{-at} - e^{-\frac{R}{L}t} \right]$$

Problem 3

Find the current $i(t)$ in an LC-circuit, given $L = 1$ henry, $C = 1$ farad, zero initial current, and the charge on the capacitor is

$$v(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Solution: In an LC circuit (i.e. in LCR, $R = 0$), the differential equation is

$$L \frac{d^2 Q}{dt^2} + \frac{1}{C} Q = E(t) \Rightarrow \frac{d^2 Q}{dt^2} + Q = \begin{cases} t, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad (\because L = 1, C = 1) \quad (1)$$

Apply the L.T. on (1):

$$\left[s^2 Q(s) - s Q(0) - Q'(0) \right] + Q(s) = \int_0^1 t e^{-st} dt + \int_1^\infty 0 dt \quad (\because \text{by definition})$$

Apply $Q(0) = 0$, $Q'(0) = 0$ and proceed similarly to complete the solution.

17 Application in Mechanical Vibrations

Beam deflection problem

A load P acts downward through the portion $0 \leq x \leq a$ of a beam of length ℓ ; find the deflection $y(x)$.

Solution: We know that the deflection of the beam $y(x)$, based on the given conditions, is given by

$$y^{(iv)}(x) = k \delta(x - a) \quad (1)$$

with boundary conditions

$$y(0) = y''(0) = y(\ell) = y''(\ell) = 0.$$

To apply the L.T. on (1), we need two more boundary conditions $y'(0)$ and $y'''(0)$. Let us assume

$$y'(0) = a, \quad y'''(0) = b.$$

Now apply the L.T. on (1):

$$\begin{aligned} \mathcal{L}(y^{(iv)}(x)) &= k \mathcal{L}(\delta(x - a)) \\ \Rightarrow \left[s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) \right] &= k e^{-as} \end{aligned}$$

Now apply the boundary conditions:

$$\begin{aligned} \Rightarrow s^4 Y(s) - as^2 - b &= k e^{-as} \\ \Rightarrow s^4 Y(s) &= k e^{-as} + as^2 + b \\ Y(s) &= k \left\{ \frac{e^{-as}}{s^4} \right\} + \frac{a}{s^2} + \frac{b}{s^4} \end{aligned}$$

Apply the inverse L.T. on both sides:

$$\begin{aligned} \Rightarrow y(x) &= k \left\{ \mathcal{L}^{-1} \left(\frac{e^{-as}}{s^4} \right) \right\} + a \mathcal{L}^{-1} \left(\frac{1}{s^2} \right) + b \mathcal{L}^{-1} \left(\frac{1}{s^4} \right) \\ &= k \left[\mathcal{L}^{-1} \left(\frac{1}{s^4} \right) \right]_{x \rightarrow x-a} u(x-a) + ax + b \frac{x^3}{3!} \\ &= k \left(\frac{x^3}{3!} \right)_{x \rightarrow x-a} u(x-a) + ax + \frac{bx^3}{6} \end{aligned}$$

$$y(x) = \frac{k}{6} (x-a)^3 u(x-a) + ax + \frac{bx^3}{6}$$

18 Solution of Integral Equations using L.T.

Problem 1

Solve $y(t) = t^2 + \int_0^t y(u) \sin(t-u) du$.

Solution: Apply the L.T. on the above equation:

$$\begin{aligned}\mathcal{L}(y(t)) &= \mathcal{L}(t^2) + \mathcal{L}\left(\int_0^t y(u) \sin(t-u) du\right) \\ \Rightarrow Y(s) &= \frac{2}{s^3} + \mathcal{L}(y(t) * \sin t) \quad \left(\because f(t) * g(t) = \int_0^t f(u) g(t-u) du, \text{ convolution defn.}\right) \\ &= \frac{2}{s^3} + \mathcal{L}(y(t)) \cdot \mathcal{L}(\sin t) \quad (\because \mathcal{L}(f(t) * g(t)) = F(s) \cdot G(s)) \\ Y(s) &= \frac{2}{s^3} + Y(s) \frac{1}{s^2 + 1} \\ \Rightarrow Y(s) \left(1 - \frac{1}{s^2 + 1}\right) &= \frac{2}{s^3} \\ \Rightarrow Y(s) \left(\frac{s^2 + 1 - 1}{s^2 + 1}\right) &= \frac{2}{s^3} \\ \Rightarrow Y(s) \left(\frac{s^2}{s^2 + 1}\right) &= \frac{2}{s^3} \\ \Rightarrow Y(s) &= \frac{2}{s^3} \times \left(\frac{s^2 + 1}{s^2}\right) \\ &= 2 \left(\frac{s^2 + 1}{s^5}\right) \\ Y(s) &= 2 \left(\frac{1}{s^3} + \frac{1}{s^5}\right)\end{aligned}$$

Take the inverse L.T.:

$$\begin{aligned}y(t) &= 2 \left\{ \mathcal{L}^{-1}\left(\frac{1}{s^3}\right) + \mathcal{L}^{-1}\left(\frac{1}{s^5}\right) \right\} \\ &= 2 \left\{ \frac{t^2}{2!} + \frac{t^4}{4!} \right\} \\ &= \frac{t^2}{1} + \frac{t^4}{12}\end{aligned}$$

$$y(t) = \frac{12t^2 + t^4}{12}$$

Exercise: Problem 2

Solve $y + \int_0^t y dt = 1 - e^t$.

19 Exercises**I. Find the L.T. of the following**

- (1) $\cos \pi t + 4e^{2t/3}$
- (2) $a + bt + \frac{c}{\sqrt{t}}$
- (3) $4^t + |t|$
- (4) $|t - 1| + |t + 1|, \quad t \geq 0$
- (5) $e^{-t} \cos t \sin t$
- (6) $t^3 e^{-3t}$
- (7) $t^2 \sin at$
- (8) $\frac{1 - \cos t}{t^2}$
- (9) $\sin \sqrt{t}$ and hence find $\frac{\cos \sqrt{t}}{\sqrt{t}}$
- (10) $t^2 u(t - 3)$
- (11) $\int_0^t e^{-t} \cos t dt$

II. Evaluate using L.T.

- (1) $\int_0^\infty \left(\frac{\cos at - \cos bt}{t} \right) dt$
- (2) $\int_0^\infty \frac{e^{-t} \sin^2 t}{t} dt$

III. Find $\mathcal{L}^{-1}$ of the following

- (1) $\frac{s - 2}{6s^2 + 20}$
- (2) $\frac{14s + 10}{49s^2 + 28s + 13}$
- (3) $\frac{s}{4s^2 - 25}$
- (4) $\frac{s^2 + 3}{s(s^2 + 9)}$
- (5) $\frac{s e^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}$

$$(6) \log_e \left(\frac{s^2 - 1}{s^2} \right)$$

$$(7) \tan^{-1} \left(\frac{2}{s^2} \right)$$

$$(8) \cot^{-1} \left(\frac{s+3}{2} \right)$$

$$(9) \frac{s^2 - a^2}{(s^2 + a^2)^2}$$

$$(10) \frac{s^3}{s^4 - a^4}$$

$$(11) \frac{5s + 3}{(s - 1)(s^2 + 2s + 5)}$$

$$(12) \frac{1}{s^3(s^2 + 1)} \quad (\text{use Convolution})$$

IV–VI. Word problems

IV. A particle moves in a line so that its displacement x from a fixed point O at any time t is given by

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 50 \sin 5t.$$

Using L.T., find its displacement at any time t if initially the particle is at rest at $x = 0$.

V. Solve $[tD^2 + (1 - 2t)D - 2]y = 0$, given $y(0) = 1$ and $y'(0) = 2$.

VI. The coordinates (x, y) of a particle moving along a plane curve at any time t are given by

$$\frac{dy}{dt} + 2x = \sin 2t, \quad \frac{dx}{dt} - 2y = \cos 2t.$$

It is given that at $t = 0$, $x = 1$ and $y = 0$. Show that, using the L.T. technique, the particle moves along the curve $4x^2 + 4xy + 5y^2 = 4$.